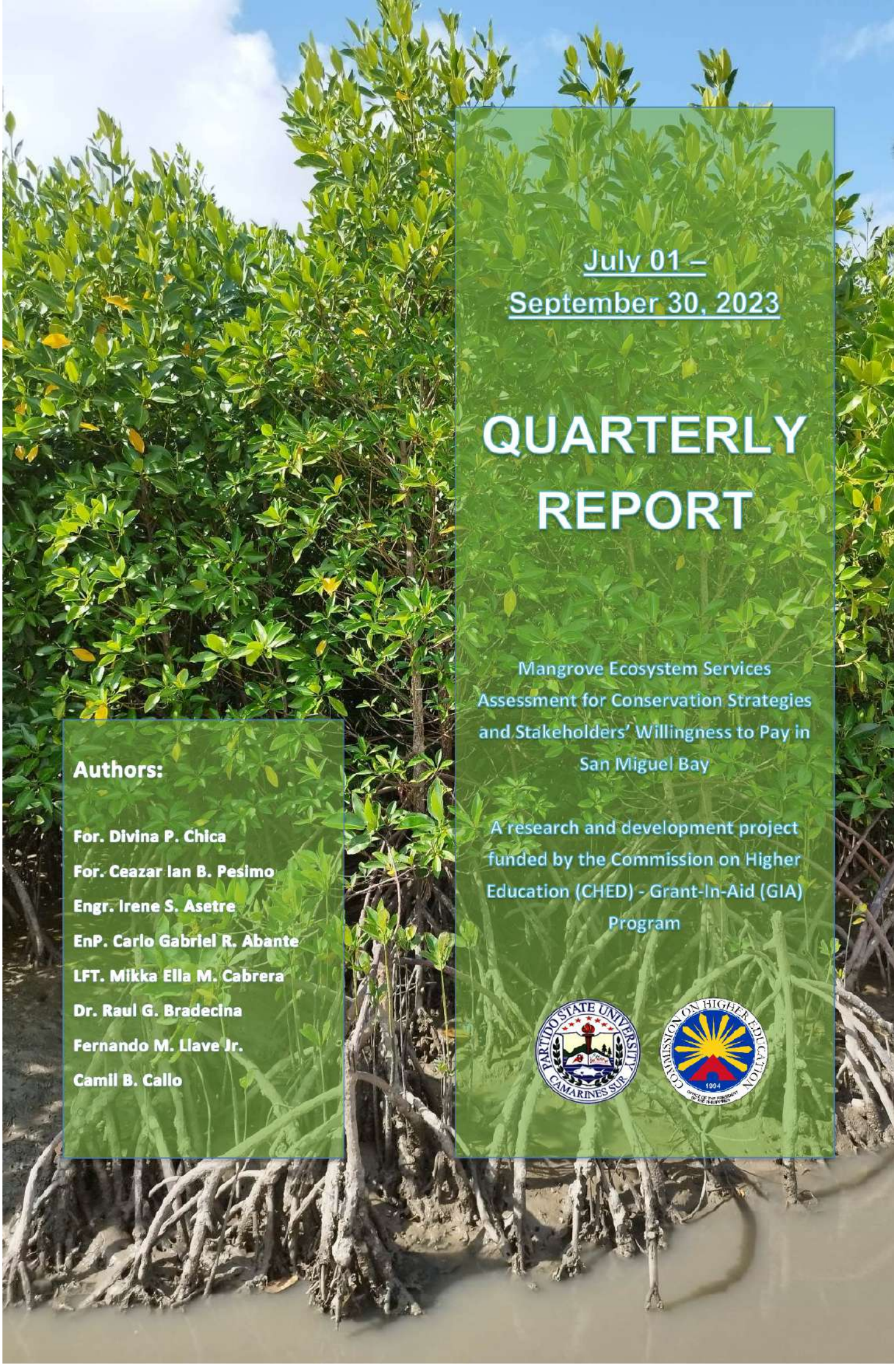


July 01 –
September 30, 2023

QUARTERLY REPORT

Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay

A research and development project funded by the Commission on Higher Education (CHED) - Grant-In-Aid (GIA) Program

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About the Project

Project Title:

Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay

Component Studies

1. Community Structure and biodiversity Assessment of Mangrove Ecosystem in San Miguel Bay
2. Inventory of Ecosystem Services of Mangrove Forest and Narrative of the Benefits to Coastal Communities of San Miguel Bay
3. Assessment of Fisheries Production of Mangrove Ecosystem in San Miguel Bay
4. Contingent Valuation Study on Mangrove Forest Ecosystem Conservation and the fisher folks and coastal communities in San Miguel Bay
5. Socio-psychological Factors affecting Utilization and Willingness to Participate in the Management and Conservation of Mangrove Ecosystem of San Miguel Bay

Project Members

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Project Activities

This quarterly report provides a comprehensive overview of the research activities conducted by the research team during the period from July to September 2023. It aims to summarize the key findings, progress, and outcomes achieved throughout the specified timeframe. This report facilitates effective communication, evaluation, and strategic planning for future endeavors.

In this case, the project activities for the project as scheduled are the following:

- Implementation Progress
- Administrative Support Activities
    - Preparation of materials and equipment for fieldwork activities
    - Status of Memorandum of Agreements (MOA) with three (3) local government units (LGUs) surrounding San Miguel Bay (SMB), Municipal Clearances and Ethics Clearance for Questionnaires
    - Preparation of Survey Questionnaires and Monitoring of Enumerators
    - Preparation of Quarterly Report (Q3)
- Research Undertakings
    - Presentation of Research Project
    - Household Surveys
    - Mangrove Forest Inventory
    - Fisheries Assessment

Implementation Progress

The implementation progress of the research project has been steadily advancing during the reporting period.

Administrative Support Activities

The administrative support activities in this project, for the period of July to September 2023, includes the following: (1) preparation of materials and equipments for fieldwork, (2) updating the status of Memorandum of Agreements (MOA) with three (3) localities surrounding SMB and securing of Municipal Clearances, and Ethics Clearance from University Ethics Committee; preparation of survey questionnaires and monitoring of enumerators; and (3) preparation of Quarterly Report (Q3).

During the 3rd Quarter of implementation of this project, the project staff and project support staff worked together to prepare the materials and equipment for fieldwork activities. This includes the preparation of survey questionnaires for household surveys, monitoring the targets and accomplishments of enumerators, mangrove tags and other field equipment for mangrove forest inventory and fisheries assessment.

In terms of necessary documents, project support staff facilitated the communication with three (3) LGUs regarding the status of MOAs and municipal clearances. Also, constant

communication to the University Ethics Committee regarding the status of request for ethics clearance for survey questionnaires.

By the end of the 3rd quarter, reports have been prepared summarizing the overall conducted activities.

Research Undertakings

In every research task, achieving efficiency relies on proper preparation and the adoption of a methodical approach. Starting July 1, 2023 until September 30, 2023, the following are the tasks which are accomplished by the project team, relative to the approved timeline of the project i.e., presentation of research project; household surveys; mangrove forest inventory; and fisheries assessment. The following sections will be discussing the details of the mentioned activities, including photo documentation and research outputs.

Memorandum of Agreement (MOA), Clearances, and Permits

Table 1 shows the status of MOA, clearances and permits. Four (4) out of seven (7) LGUs have already signed a MOA, two (2) issued an SB resolution and LGU Cabusao still for SB resolution. In terms of municipal clearance, three (3) out of seven (7) LGUs issued a clearance (**Appendix**). Two (2) requested prior informed clearance for critical habitats were granted (**Appendix**) and two (2) gratuitous permits have been granted by DENR.

Table 1. Status of MOA, Clearances, and Permits

LGU	MOA	Clearance	Permit	Remarks
Siruma	*Issued SB resolution*	*On process*	N/A	• SB Resolution have been issued and ready for MOA signing
Tinambac	✓	*On process*	N/A	
Calabanga	*Issued SB resolution*	✓	N/A	• SB Resolution have been issued and ready for MOA signing
Sipocot	✓	*On process*	N/A	
Cabusao	*For SB resolution*	✓	Prior Informed Clearance: ✓ Gratuitous Permit: ✓	
Basud	✓	*On process*	N/A	
Mercedes	✓	✓	Prior Informed Clearance: ✓ Gratuitous Permit: ✓	

Presentation of Research Project

The research team aims to present the research project to the Municipal Fisheries and Aquatic Resource Management Councils (MFARMCs) of seven (LGUs) surrounding San Miguel Bay. This is to give the councils a background of the study and the expected outputs of the research project. The project also aims to gather secondary data from different councils.

On August 18, 2023, the research team presented the research study to the MFARMC of LGU Siruma. They introduced the members of the research team and provided a brief overview of the study, including its scope and objectives. They also presented an overview of their methods for conducting mangrove forest inventory and fisheries assessment. Following the presentation, the council provided valuable feedback and recommendations to the research team. These included requesting access to existing mangrove assessment data in Siruma, exploring mangrove areas in other barangays, and engaging with People's Organization (PO) working on mangrove management in the municipality. Similarly, the research study was also presented to the MFARMC of LGU Mercedes on September 27, 2023. The council emphasized the need to minimize the disturbances to mangrove areas, suggested requesting the existing mangrove data from the Municipal Agriculture Office (MAO), and offered their assistance during mangrove forest inventory and fisheries assessment.



Figure 1. Project Team from Partido State University - Tinambac Campus, during the presentation of research study to Municipal Fisheries and Aquatic Resource Management Council (MFARMC) meeting in LGU Siruma and Mercedes.

Household Survey

Survey questionnaires of components 2, 3, 4, and 5 were distributed to the enumerators. Household surveys were started in the month of August 2023.

Mangrove Forest Inventory

Mangrove Forest Inventory conducted during the Third Quarter of Year 2023. Below is the summarized table of mangrove assessment in seven LGUs.

Table 2 shows the summarized mangrove assessment in seven (7) LGUs. The assessment was conducted from the month of July to September 2023 in ten (10) stations covering nine (9) barangays. Thirty (30) plots were established and assessed.

Table 2. Third Quarter Mangrove Assessment

Local Government Unit	Date	Barangay	Station	Transects/Plots
Siruma	July 07, 2023 September 11, 2023	Bahao	S88	T1 - P2 T1 - P1, T2 - P1
Tinambac	July 07, 2023 July 25, 2023 September 07, 2023	Old Caaluan	S53	T1 - P3; T2 - P1 T2 - P2,P3 T3 - P1,P2
Calabanga	July 04, 2023 September 19, 2023	Punta Tarawal	S36	T1 - P1,P2 T2 - P1, P2
Cabusao	July 17, 2023 September 21, 2023	Barceloneta Biong	S25,S26 S26	T1 - P2 T2 - P1
Sipocot	July 18, 2023 September 20, 2023	Cotmo	S20,S21 S20	T2 - P2; T2 - P1 T1 - P1,P2
Basud	July 04, 2023 July 05, 2023 July 27, 2023 September 12, 2023 September 13, 2023 September 26, 2023	Taba-taba Mangcamagong Taba-taba Mangcamagong Taba-taba	S5 S7 S5 S8 S5	T1 - P1 T2 - P1 T1 - P2 T1 - P3 T1 - P3 T1 - P4
Mercedes	July 05, 2023	Matoogtoog	S8	T1 - P2 T2 - P1

Table 2 shows the initial data on mangrove forest inventory conducted in seven (7) LGUs. There are eighteen (18) mangrove species identified, namely, *Avicennia rumphiana*, *Avicennia marina*, *Avicennia officinalis*, *Bruguiera cylindrica*, *Bruguiera parviflora*, *Bruguiera sexangula*, *Bruguiera gymnorrhiza*, *Ceriops decandra*, *Ceriops tagal*, *Excoecaria agallocha*, *Heretiera littoralis*, *Lumnitzera littorea*, *Rhizophora mucronata*, *Rhizophora apiculata*, *Scyphiphora hydrophyllaceae*,

Sonneratia alba, *Sonneratia ovata*, and *Xylocarpus granatum*. Also, one (1) beach species was identified, namely, *Pongania pinnata*.

Table 3. Initial Data on Mangrove Forest Inventory in seven (7) LGUs

Common Name	Genus	Specific epithet
1. Miapi/Kuyapi	*Avicennia*	*rumphiana*
2. Bungalon	*Avicennia*	*marina*
3. Miapi/Kuyapi	*Avicennia*	*officinalis*
4. Pototan/Busain	*Bruguiera*	*cylindrica*
5. Hangarai	*Bruguiera*	*parviflora*
6. Pototan	*Bruguiera*	*sexangula*
7. Pototan	*Bruguiera*	*gymnorrhiza*
8. Baras-baras	*Ceriops*	*decandra*
9. Tangal	*Ceriops*	*tagal*
10. Lipata/Buta-buta	*Excoecaria*	*agallocha*
11. Dungon	*Heretiera*	*littoralis*
12. Culasi	*Lumnitzera*	*littorea*
13. Bakawang Lalaki	*Rhizophora*	*apiculata*
14. Bakawang Babae	*Rhizophora*	*mucronata*
15. Nilad	*Scyphiphora*	*hydrophyllacea*
16. Pagatpat	*Sonneratia*	*alba*
17. Tabigi	*Xylocarpus*	*granatum*
18. Piag-ao	*Xylocarpus*	*moluccensis*
19. Anonas (Beach sp.)	*Annona*	*squamosa*
20. Putat laut (Beach sp.)	*Barringtonia*	*asiatica*
21. Dapdap (Beach sp.)	*Eryhtrina*	*variegata*
22. Barok-barok (Beach sp.)	*Fabaceae*	*sp.*
23. Malubago (Beach sp.)	*Hibiscus*	*tiliaceus*
24. Talisay (Beach sp.)	*Terminalia*	*catappa*

Photo Documentation



Figure 2. Project Team from Partido State University - Tinambac Campus, mangrove assessment undertaken during the Third Quarter of the year.

Fisheries Assessment

Fisheries Assessment conducted during the Third Quarter of Year 2023. Below is the summarized table of fisheries assessment in seven LGUs.

Table 4. Third Quarter Fisheries Assessment

Identification of Fish and Macro-invertebrates associated in Mangrove Areas of San Miguel Bay								
SPECIES		MUNICIPALITY						
English name	Scientific Name	Siruma	Tinambac	Calabanga	Cabusao	Sipocot	Basud	Mercedes
	Batillaria spp.							
	Cerithidea spp.							
	Hermit crab							
	Nerita spp. 1							
Cockle	*Anadara spp.*							
Kapis	*Isognomon sp.*							
Talaba	*Crassostrea spp*							
	Ocenabra sp.							
	Potamides sp.							
	Pirenella							
orange fiddler	*Tubuca coarctata*							
Mangrove tree crab	*Aratus pisonii*							
Spotted moon crab	*Ashtoret lunaris*							
Mud crab	*Scylla sp.*							
Ring-legged crab	*Austruca annulipes*							
	Austruca triangularis							
	Gelasimus jocelynae							
Yellow-orange crab								
Shrimp	*Metapenaeus sp.*							
Snapping shrimp	*Alpheus sp.*							
Duckbill	*Linguna anatina*							
Venus clam	*Chamelea sp.*							
Manila clam								
Goby								
Halfbeak								
Hardy silverside	*Atherinomorus sp.*							
Sea krait								
Stripped catfish								

Gray color- Present

White color- Not present during sampling

Photo Documentation

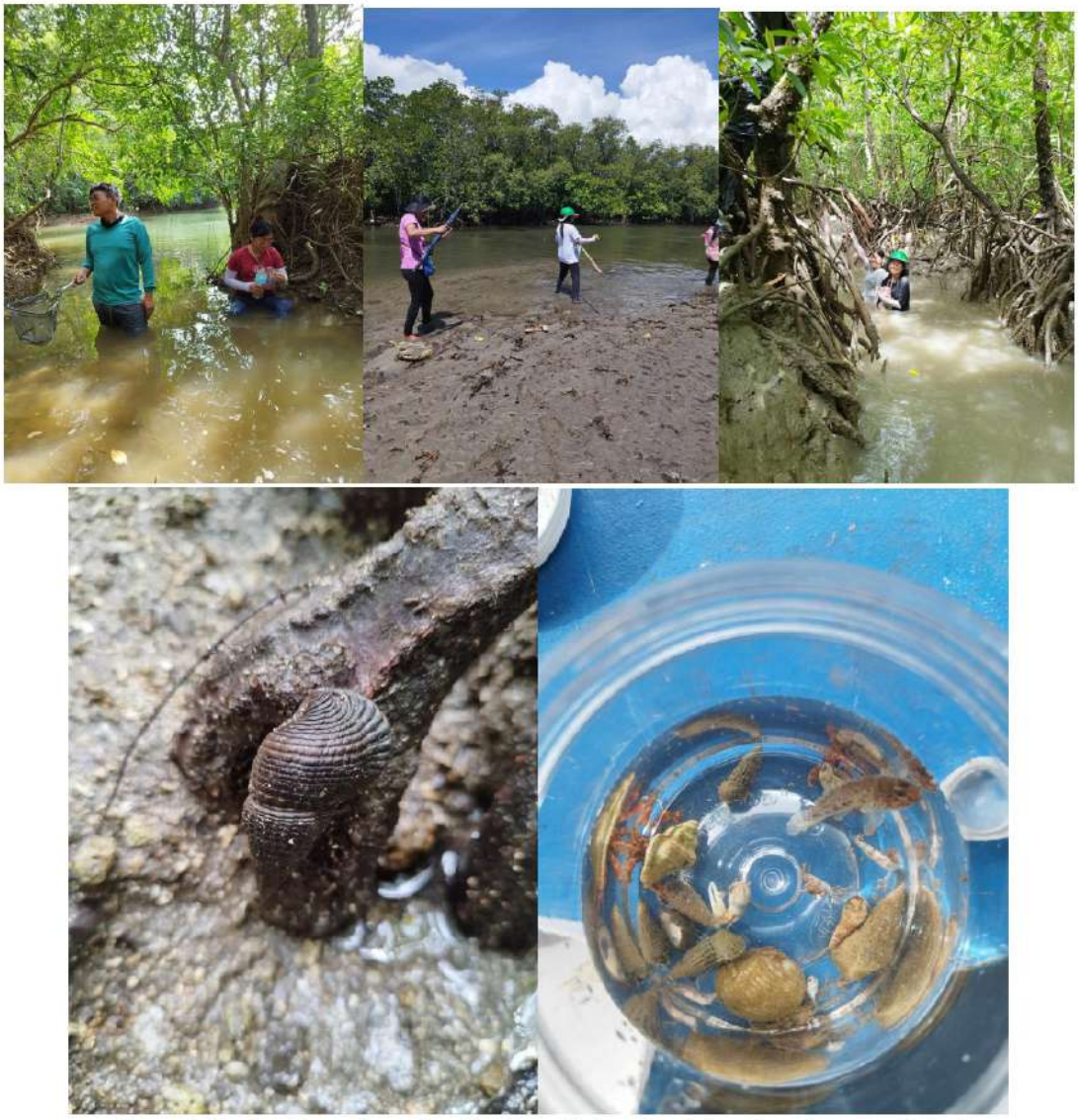


Figure 3. Project Team from Partido State University - Tinambac Campus, fisheries assessment undertaken during the Third Quarter of the year.

Financial Status

The project was granted a total research fund of PHP 4,500,000.00 for two (2) years (September 16, 2022 to September 15, 2024). In this line, a total of 1,938,242.00 was requested for Year-2023 (January to December 2023) expenses. As of September 30, 2023, the estimated budget utilization during this year is PHP 1,076,874.68 (**Table 5**).

Table 5. Financial Status of the Mangrove Research Project of ParSU-Tinambac as of September 30, 2023, funded by CHED-GIA.

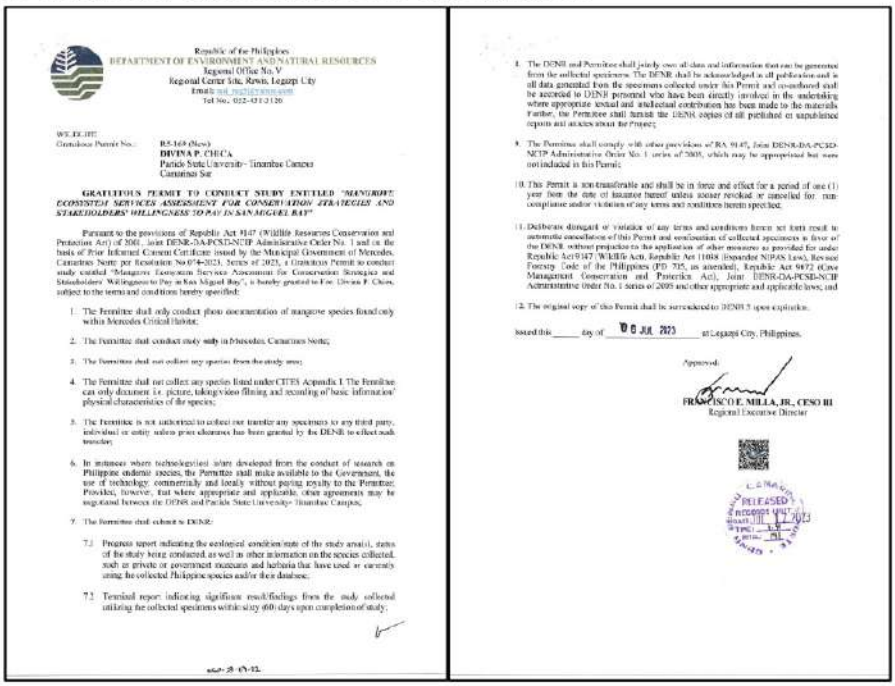
Partido State University		
Agency		
FUND UTILIZATION REPORT		
Project: Research Innovation and Extension Proposal: Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay		
As of September 30, 2023		
Beginning Balance	-	
Funds Received:NCA-ROV-23-0001228 dated March 17, 2023	1,670,984.00	**1,670,984.00**
Funds Utilized		
I.Personnel Services (PS)		
Personnel Services		
Honoraria (Project Leader & Members)	346,600.00	
Sub- Total		346,600.00
II. Maintenance and Other Operating Expenses (MOOE)		
Travel Expenses	187,230.00	
Supplies and Materials	142,275.38	
Communication Expenses	-	
Professional Services		
COS-Project Support Staffs	320,069.30	
Enumerators' Fee	-	
Other MOOE		
Printing and Publication Expenses	-	
Representation Expenses	-	
Subscription Expenses	-	
Sub-Total		649,574.68
III. Admin Cost (MOOE and Honoraria for Admin Support Staffs)		10,700.00
IV. Equipment Outlay		70,000.00
TOTAL		1,076,874.68
Ending Balance (Total Funds Received - Total Funds Utilized)		**594,109.32**

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APPENDICES

Appendix 1. Gratuitous Permit

Appendix 1.1 Gratuitous Permit for Mercedes Critical Habitat



Republic of the Philippines
DEPARTMENT OF ENVIRONMENT AND NATURAL RESOURCES
Regional Office No. V
Regional Center Site, Rawis, Legazpi City
Email: [redacted]
Tel No.: 052-481-2120

WILDLIFE
Gratuitous Permit No.: R5-169 (New)

DIVINA P. CHICA
Partido State University- Tinambac Campus
Camarines Sur

GRATUITOUS PERMIT TO CONDUCT STUDY ENTITLED *"MANGROVE ECOSYSTEM SERVICES ASSESSMENT FOR CONSERVATION STRATEGIES AND STAKEHOLDERS' WILLINGNESS TO PAY IN SAN MIGUEL BAY"*

Pursuant to the provisions of Republic Act 9147 (Wildlife Resources Conservation and Protection Act) of 2001, Joint DENR-DA-PCSD-NCIP Administrative Order No. 1 and on the basis of Prior Informed Consent Certificate issued by the Municipal Government of Mercedes, Camarines Norte per Resolution No.07-6-2023, Series of 2023, a Gratuitous Permit to conduct study entitled "Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay", is hereby granted to Dr. Divina P. Chica, subject to the terms and conditions hereby specified:

1. The Permittee shall only conduct photo documentation of mangrove species found only within Mercedes Critical Habitat;
2. The Permittee shall conduct study only in Pinagsaka, Camarines Norte;
3. The Permittee shall not collect any species from the study area;
4. The Permittee shall not collect any species listed under CITES Appendix I. The Permittee can only document i.e. picture, taking video filming and recording of basic information' physical characteristics of the species;
5. The Permittee is not authorized to collect nor transfer any specimens to any third party, individual or entity unless prior clearance has been granted by the DENR to effect such transfer;
6. In instances where technology/ies) is/are developed from the conduct of research on Philippine endemic species, the Permittee shall make available to the Government, the use of technology commercially and locally without paying royalty to the Permittee; Provided, however, that where appropriate and applicable, other agreements may be negotiated between the DENR and Partido State University- Tinambac Campus;
7. The Permittee shall submit to DENR:
   7.1 Progress report indicating the ecological condition/state of the study area(s), status of the study being conducted as well as other information on the species collected, such as private or government museums and herbaria that have used or currently using the collected Philippine species and/or their data/use;
   7.2 Terminal report indicating significant result/findings from the study collected utilizing the collected specimens within sixty (60) days upon completion of study;

8. The DENR and Permittee shall jointly own all data and information that can be generated from the collected specimens. The DENR shall be acknowledged in all publication and in all data generated from the specimens collected under this Permit and co-authored shall be accorded to DENR personnel who have been directly involved in the undertaking where appropriate textual and intellectual contribution has been made to the materials. Further, the Permittee shall furnish the DENR copies of all published or unpublished reports and studies about the Project;
9. The Permittee shall comply with other provisions of RA 9147, Joint DENR-DA-PCSD-NCIP Administrative Order No. 1 series of 2005, which may be appropriate but were not indicated in this Permit;
10. This Permit is non-transferable and shall be in force and effect for a period of one (1) year from the date of issuance hereof unless sooner revoked or cancelled for non-compliance and/or violation of any terms and conditions herein specified;
11. Deliberate disregard or violation of any terms and conditions herein set forth result to automatic cancellation of this Permit and confiscation of collected specimens in favor of the DENR without prejudice to the application of other measures as provided for under Republic Act 9147 (Wildlife Act), Republic Act 11038 (Expanded NIPAS Law), Revised Forestry Code of the Philippines (PD 705, as amended), Republic Act 9072 (Cave Management Conservation and Protection Act), Joint DENR-DA-PCSD-NCIP Administrative Order No. 1 series of 2005 and other appropriate and applicable laws; and
12. The original copy of this Permit shall be surrendered to DENR 5 upon expiration.

Issued this _____ day of 06 JUL 2023 at Legazpi City, Philippines.

Approved:

FRANCISCO E. MILLA, JR., CESO III
Regional Executive Director

RELEASED